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| 100W USB Type-C Power Bank | Dual USB Power Bank | PD Power Bank |
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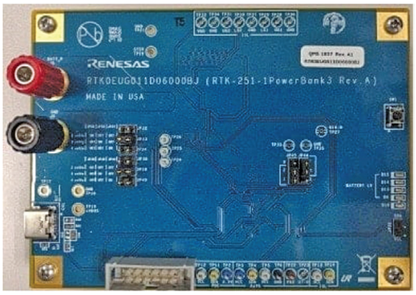
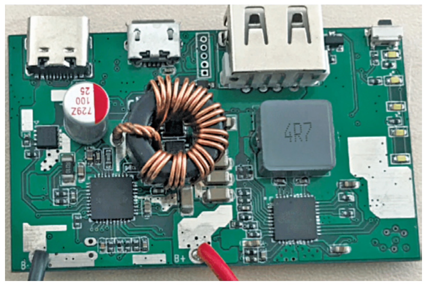
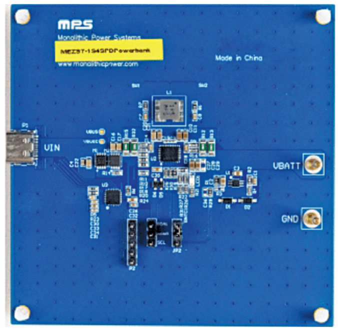
|  |  |  |
| This USB-certified power bank features adjustable cell configurations, USB Type-C connectivity, and programmable power supply options, ensuring adaptable, safe, and efficient device charging. | The portable power bank design offers dual USB functionality and advanced power delivery, with comprehensive protection systems and robust performance across extreme temperatures. | The power bank design supports PD3.0 and BC1.2 protocols with a flexible voltage range, optimised for devices using USB power delivery standards. |
| Renesas has introduced a USB-certified single-port power bank, optimised for lithium-ion batteries and supporting configurations of 2 to 4 cells in series with a default setting of 3 cells. This device features a USB Type-C port with dual role power (DRP) functionality, allowing it to alternate between sink mode for charging itself and source mode to power external devices; the source mode also supports a programmable power supply (PPS) for varied applications. The power bank is customisable through standard firmware that can be adjusted to accommodate different cell configurations, enhancing its adaptability. A firmware generation support tool with a graphical user interface allows users to modify USB PD protocol parameters, granting more control over functionality and performance. Safety is bolstered by protection mechanisms for over-temperature at the USB Type-C receptacle, over-voltage, and current on VBUS, ensuring the device's safe and reliable operation. | Infineon Technologies' CCG3PA power bank reference design features a portable external battery that stores and delivers charge through a USB Type-C receptacle that functions as both a power source and a sink, termed a dual role port. It supports Type-C and USB power delivery (PD) for USB PD Revision 3.1, which includes programmable power supply (PPS) mode. It offers protection including VBUS to CC short protection, overvoltage, overcurrent, under-voltage, short circuit protection, and overtemperature protection through an ADC. The design features a 32-bit MCU subsystem with an ARM Cortex-M0 CPU, 64kB Flash, and 8kB SRAM, and includes clocks for operations ranging from 3.0V to 24.5V (30V tolerant). Additional features include ESD protection covering various pins to meet IEC61000-4-2 level 4C standards and packaging options like 24-pin QFN and 16-pin SOIC to ensure performance across a temperature range from -40°C to +105°C. | The MEZZ 7-1S-4SPDPowerBank from Monolithic Power Solution is a reference design featuring the MP2651 buck-boost charger and the CCG3PA PD controller from Cypress, tailored for PD applications. This module supports PD3.0 and BC1.2 protocols, catering to devices adhering to USB power delivery standards. The charger accommodates 1 to 4 cell battery packs, with a flexible input voltage range of 3V to 21V at the source mode input pin, aligning with USB PD specifications. The layout incorporates commonly used capacitors, with optimised charging settings for a 2-cell Li-ion battery set at 5A and 8.4V. In reverse source mode, it outputs 5V at 3A. This design is well-suited for various industry professionals including engineers, product designers, device manufacturers, battery pack producers, and power management solution providers, as well as educational institutions where it can aid in the study and innovation of power delivery mechanisms. |
| The reference design is suitable for various applications, enhancing its versatility and usability. | The reference design is suitable for USB Type-C port and one Type-A port charging | The reference design is suitable for PD applications. |
| Renesas | Infineon Technologies | Monolithic Power Solution (MPS) |
| https://www.electronicsforu.com/electronics-projects/usb-type-c-power-bank-reference-design | https://www.electronicsforu.com/electronics-projects/power-bank-reference-design | https://www.electronicsforu.com/electronics-projects/pd-power-bank-reference-design |